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## Radiation studies performed on the High Luminosity ATLAS TileCal link Daughterboard

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on behalf of the ATLAS Tile Calorimeter System

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**ABSTRACT:** The new electronics of the ATLAS Tile Calorimeter for the HL-LHC interfaces the on-detector and off-detector electronics by means of a Daughterboard. The Daughterboard is positioned on-detector featuring commercial SFPs+, CERN GBTx ASICs, ProASIC FPGAs and Kintex Ultrascale FPGAs. The design minimizes single points of failure and mitigates radiation damage by means of a double redundant scheme, Triple Mode Redundancy, Xilinx Soft Error Mitigation IP, CRC/FEC for link data transfer, and SEL protection circuitry. We present an updated summary of the TID, NIEL and SEE qualification tests, and performance studies of the Daughterboard revision 6 design.

**KEYWORDS:** Radiation damage to electronic components; Radiation-hard detectors; Radiation-hard electronics

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